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First results from the interaction between hydroxyl radicals and condensed water droplets: A spin-polarized DFT-MD study

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ABSTRACT

The formation of clouds results from condensation of water vapor into liquid water droplets. These water droplets form on the surface of tiny particles that are suspended in the air, like dust. Atmospheric chemistry relies heavily on the hydroxyl radical (OH*), which is highly reactive and prone to combining with hydrogen atoms in other compounds. Despite previous studies concerning liquid, solid, and gas phase interactions, how hydroxyl radicals interact with condensed water droplets is still unknown. OH*-condensed droplet interactions show unique behavior, according to our studies. Several hydrogen transfers were detected during the interaction process. It takes approximately 3.5 kcal/mol of energy to initiate hydrogen transfer. Moreover, we revealed the dominant structures of hydrated radicals during the mobility of OH* into condensed droplets using Born–Oppenheimer molecular dynamics simulations. In contrast to our previous studies, the hemibond doesn't significantly affect the stability of $OH^*(H_2O)_n$ structures in condensed droplets. Our results showed that condensed droplets hinder the mobility of OH*, causing its diffusion coefficient to decrease. Moreover, several analyses were carried out to investigate condensed droplet surfaces, bonding patterns, and the position of OH* in conjunction with water molecules, including mobility analyses, radial distribution functions, coordination analyses, and population analyses.

1. Introduction

There is no denying the importance of clouds to the Earth's climate. They can shade the ground and block the sun, cooling the planet's atmosphere. The chemical reactions within clouds can be complex and vary depending on the specific composition of the cloud and the surrounding atmosphere. For example, reactions between water vapor and some atmospheric species such as nitrogen oxides and sulfur dioxide can produce nitric acid and sulfuric acid, contributing to acid rain and other environmental damage [1]. Clouds also trap heat and humidity in the atmosphere, warming the air. Based on several field campaigns, a substantial fraction of cloud droplets is activated on ultrafine aerosol particles with a diameter of less than 0.1 mm, particularly in clean or moderately polluted conditions [2–4]. Aerosol particles vary in size, composition, and location, ranging from tiny clusters of a few molecules to tens of micron-sized cloud droplets. The formation of clouds results from condensation of water vapor into liquid water droplets. These water droplets form on the surface of tiny particles in the air that are suspended in the air, such as salt crystals, dust, and bacteria. Air-liquid droplet interfaces offer an important site for adsorption and reactions, which may accelerate atmospheric reactions or create new mechanisms [5,6]. Due to the limitations of traditional spectroscopic methods, it wasn't easy to study atmospheric reactions on aerosol water droplets. Several techniques have recently been developed to study atmospheric compounds' adsorption behavior on aerosol surfaces, including X-ray photoelectron spectroscopy, sum-frequency generation, and second-harmonic generation [7,8]. However, by using current experiment methods, it can be challenging to detect the behavior of some atmospheric species, especially in small sizes. Meanwhile, classical MD simulation can only describe physical adsorption at the air–water interfaces, while it cannot accurately describe chemical reactions in cloud droplets. Among the theoretical methods, Born-Oppenheimer molecular dynamics simulation (BOMD) has the advantage of recording reactions on short time scales as well as providing details about the mechanism, allowing us to gain a deeper understanding of aerosol effects on atmospheric chemistry. A wide range of atmospheric species, such as ozone,

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acids, radicals, and ions, can form hydrogen-bond complexes with water, influencing their photochemical properties as well as catalyzing atmospheric reactions [9–14]. Meanwhile, Hydroxyl radicals (OH*) are often called the "troposphere detergents" because they are involved in a variety of reactions related to the removal of organic compounds from the atmosphere, including greenhouse gases and pollutants [15–19]. Hydroxyl radicals are formed in the atmosphere through various chemical reactions such as the decomposition of hydroperoxides (ROOH), the reaction of excited atomic oxygen with water, and the reaction between electronically excited nitrogen dioxide and water vapor [20,21]. Hydroxyl radicals play a crucial role in chemical processes occurring in the earth's atmosphere because of their desire to form some complexes with other atmospheric constituents through the formation of a strong hydrogen bond. The lifetime of any gas that reacts with the OH* in the atmosphere can be accurately calculated by using the equation: $t = 1/k$ [OH*], where "k" refers to the first-order rate constant of the reaction. However, due to high reactivity, the experimental investigation of diffusion OH* in the atmosphere and prediction of its mechanism with other species is largely unknown. The average hydroxyl radical concentration in the troposphere is estimated as 10^6 molecule/cm^3 [22], while the water concentration is equal to half of the relative humidity at 298 K (6.95×10^{17} molecule/cm^3). Despite this, the atmosphere's water concentration varies widely, contributing to the dynamic processes taking place there. Generally, during the interaction between water and other species, such as radicals, there are four possibilities: H-bonds, in which the water acts as either a donor or acceptor, nonbonded and hemibond interactions [23]. Hemibond was proposed by Pauling as a 2-centre 3-electron bond (2c–3e), which is formed by the overlapping of a neutral molecule's lone pair orbital with its radical orbitals [24]. Based on the first results of ab initio molecular dynamics of the OH*.31H_2O system, Vassilev et al. found that the hydroxyl radical tends to form a hemibond, resulting from short O–O* distances of about 2.3 Å [25]. In a later study, Kusalik et al demonstrated that hemibonded structures disappear when the system enlarges from 31 water molecules to 63 water molecules [26]. However, our recent results showed that O–O* hemibond remains present even in the large OH*$(H_2O)_{191}$ systems [27,28], which is in accordance with the experimental results [29]. There has always been a lot of attention focused on the penetration of atmospheric species in bulk liquid water or their tendency to remain on the air–water interface. As a model for simulating the behavior of cloud droplets, Ravishankara et al. showed how a water droplet takes up a gas phase molecule [30]. In this model, different steps have been considered, including diffusion to the surface in the gas phase, surface accommodation, diffusion in the liquid media, substrate or dissolved reactant reaction, and finally, the resultant products diffuse to the surface and are expelled from the aerosol. Where the accommodation of the species at the surface has been found to be the most important step of the process. Moreover, our previous studies on water droplets revealed that hydroxyl radicals interact with water molecules at the droplet interface differently than liquid, solid, or gas phases [27,28,31]. However, in terms of aerosol collision and its role in atmospheric chemical processes at the surface, several issues remain unsolved. For example, no experimental or theoretical research has been done on the interaction of hydroxyl radicals with water molecules during the collision process. Collision occurs when cloud droplets bump into each other due to air currents within the cloud, also coalescence can occur if they collide and stick together [32]. Indeed, a major contributor to the process of collision in the air is the condensation of water droplets, which may also influence the way in which water molecules interact with OH* in the atmosphere. The goal of this paper is to explore how condensed droplets impacts on the mobility mechanism of the OH* during interaction with water nanodroplets by performing two extensive DFT-based MD simulations. Indeed, as a result of some novel H-transfer mechanisms observed during collision processes, we provided new insights into the local structure of hydrated OH* along with detection of energy barriers. Furthermore, we investigated mobility, radial distribution functions, coordination, and population analysis to examine droplet surfaces, bonding patterns, and position of OH* combined with water molecules.

2. Computational details

To gain insight into how hydroxyl radicals interact with condensed water droplets, initial conditions were obtained by placing a single OH* in different situations. Water droplets were simulated as spherical models with a radius of 9.8 Å, including 191 water molecules and a hydroxyl radical, Fig. 1. For the accurate interpretation of results obtained from the interaction between the OH* and condensed water droplets, two separate systems have been considered. According to one system (S-I), the OH* was positioned inside of the droplet at a distance of 1.2 Å from the droplet's center of mass. In contrast, in the other system (S-II), the OH* was positioned at a distance of 8.5 Å from the droplet's center of mass. Indeed, OH* position concerning the center of mass (COM) distinguishes these two systems. To calculate the distance between droplets for simulating condensed droplets, we have taken into account the widely used face-centered arrangement. Taking into account the density of the droplet, which is approximately 0.85 amu/Å^3, we have made necessary adjustments to the length of the simulation box, setting it at approximately 22 Å.

CP2K's Quickstep module was used to perform a full set of Born-Oppenheimer molecular dynamics (BOMD) simulations [33], where the wave functions were also complemented by an auxiliary plane wave basis in addition to a Gaussian basis set [34]. Before carrying out BOMD simulations, each of the systems was fully relaxed by density functional theory (DFT) simulation through the combination of Beck's parameter hybrid functional (B) with Lee-Yang-Parr correlation functional (LYP). Moreover, all systems were double relaxed by BOMD simulation for 2500 fs to ensure that the initial equilibrium have been reached. An NVT ensemble was applied to all BOMD simulations with a Nose-Hoover Chain thermostat having a time constant of 0.1 ps. To account for the weak dispersion interaction, Grime's dispersion correction DFT-D3 function has been applied based on the zero damping [35]. An assessment of the spin-polarized interaction with water molecules was performed to account for the unpaired electron on OH*. Besides the double zeta valence polarized (DZVP) basis set [36], Goedecker-Teter-Hutter (GTH) pseudopotentials were also used to complement the DZVP basis set [37,38].

Moreover, the intermediates and transition state (TS) of the reaction between the OH*–H_2O in the gas phase have been optimized at the M06-2X/6–31++G(d,p) level using the Gaussian 16 software [39]. In addition to the zero-point energies (ZPE), intrinsic reaction coordinates (IRC) calculations were used to verify the transition state [40]. A constrained MD calculation was used to calculate the energy required for hydrogen transfers in the condensed droplet systems. The average constraint force was measured over a 4.8 ps trajectory with a 0.00025 Å fs^{-1} growth rate, allowing the R to span from −0.6 Å to 0.6 Å. The free energy profile was evaluated over the defined coordinate R by a thermodynamic integration.

3. Results and discussion

3.1. The accommodation of OH* in the condensed droplet

We have shown in previous studies that OH* tends to interact with water molecules at the air–water interface of droplets due to structural characteristics [27,28,31]. These conditions have not been studied for OH* in a condensed system, especially considering the condensed water droplets. In addition to providing information about the position of the molecules, the center of mass (COM) can also be used to study the motion and properties of the molecules within the droplet. Therefore, the distance between the OH* and the droplet's COM was evaluated for the two S-I and S-II systems to track the OH* through its interaction with water molecules during 80 ps of BOMD simulation, Fig. 2. The dotted

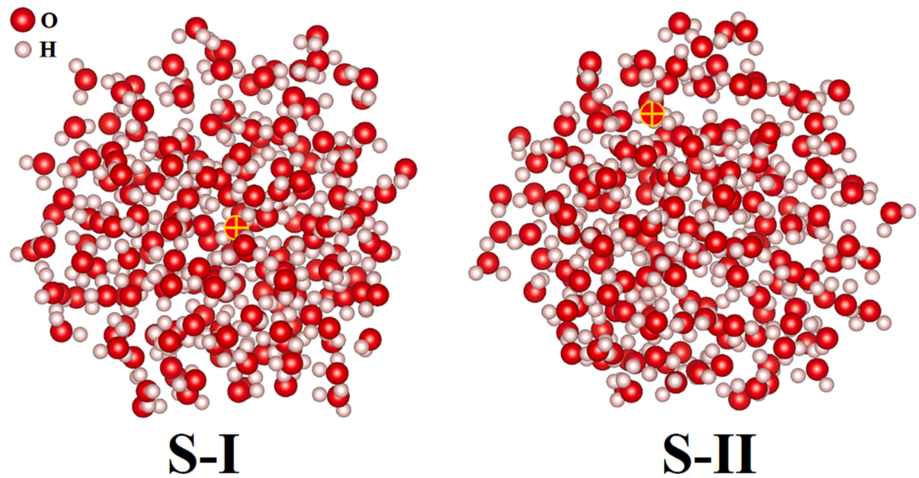

Fig. 1. Initial positions of the OH* (indicated by yellow circled plus) in the water droplet. S-I is related to the inner region (ROH = 1.2 Å) and S-II is associated with the outer region (R_{OH} = 8.5 Å). R_{OH} is defined as the distance between the OH radical and the center of mass of the droplet.

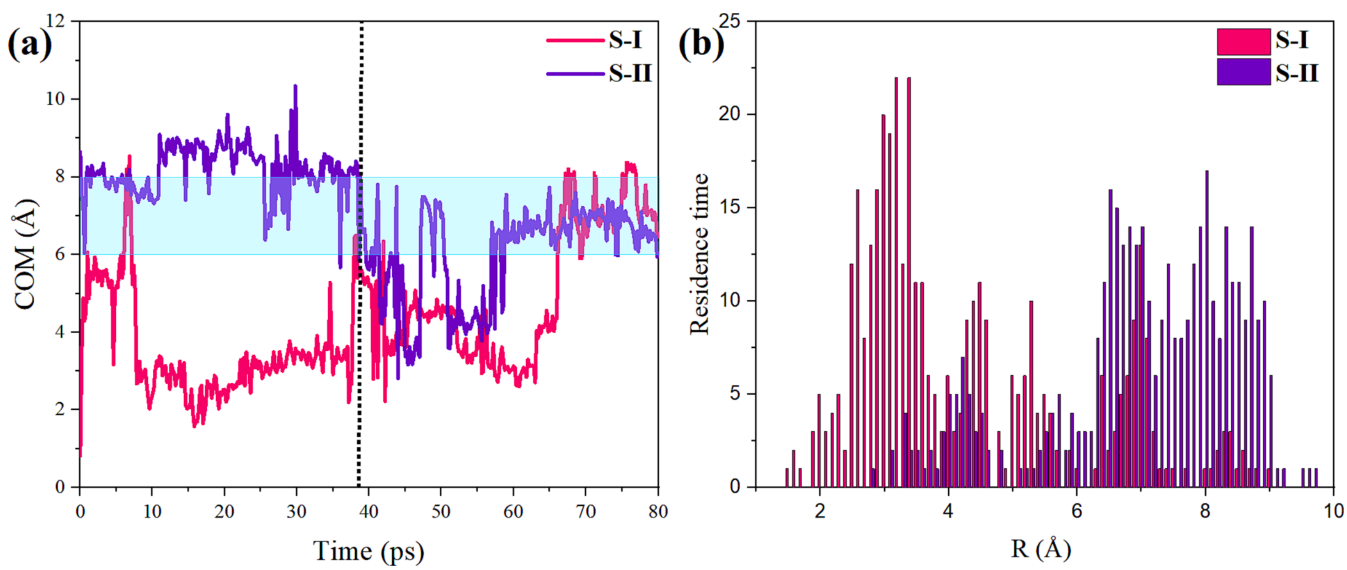

Fig. 2. (a) The distance between OH* and the center of mass (COM) of the condensed droplets during 80 ps for S-I and S-II systems. (b)The residence time distribution of OH* relative to the COM of condensed droplets for both S-I and S-II systems.

line at about 38 ps represents the moment when the distances of OH* to COM in both systems become identical, about 6 Å. Then, the OH* accommodates around 4.5 Å for about 15 ps, unsteadily. Finally, OH* in both the S-I and S-II systems reach the same stability about 7 Å from the center of mass after 68 ps and 58 ps, respectively, Fig. 2**(a)**. Additionally, the residence time distribution of OH* within the simulation time has been depicted to help better understand the OH* position within the condensed droplet. Indeed, the residence time distribution shows how long the OH* persists at a specific shell relative to COM in the water droplet. It can be found from Fig. 2**(b)** that OH* finally accommodates at 7 Å based on the common peak points of the residence time distribution in S-I and S-II systems. It should be noted that the peak points around 3 Å and 8 Å are related to the time before OH* reaches equilibrium in the S-I and S-II systems, respectively.

The density of water in the bulk is higher than the density at the interface. This means that approximately half of the water molecules in the uppermost layer of water at the air–water interface are air molecules, so the water's surface appears curved or "domed" in shape. This effect is also responsible for capillary action, where water is drawn up into narrow tubes or capillaries due to the surface tension at the air–water interface. Indeed, the droplet-air interface composition can be affected by capillary forces and the curved shape of the droplet. The capillary effect can cause hanging droplets to be bound to the air–water interface, while the curved shape of the droplet can affect its wetted diameter and wetting angle [41,42]. Upon calculating 50 % of the bulk water density, which represents the interface area [43,44], our results indicate that condensed water droplets possess an air–water interface measuring 10.5 Å, marked by a vertical dotted line in Fig. 3. Therefore, it can be concluded that the OH* does not reach the air–water interface in condensed water droplets (S-I and S-II) since it finally accommodates at 7 Å (Fig. 2). Indeed, it is likely that the collision density of condensed droplets hinders the mobility of OH*, causing it to reach an accommodation area below the air–water interface. Interestingly, our previous studies of the OH* in droplet systems at various temperatures between 200 and 300 K revealed that the OH* has a high tendency to reach the air–water interface [27,28].

In many chemical reactions, particularly those involving hydroxyl radicals, hydrogen transfer is essential in determining reaction mechanisms and kinetics. The results of DFT calculations related to OH*–H_2O in the gas phase revealed two possible pathways in which the

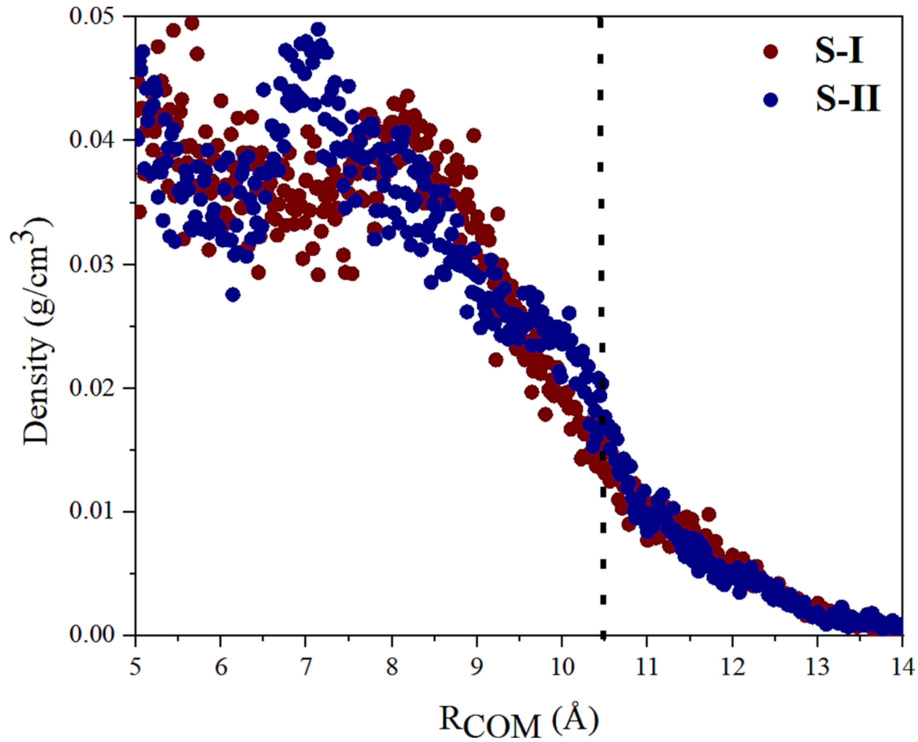

Fig. 3. The density profile of water molecules as a function of distance from the droplet's center of mass. A vertical dotted line at 10.5 indicates 50 % of bulk water density, defined as the air–water interface.

discrepancy is different orientations of the water molecule, Fig. 4. During the calculations, we found that the energy barrier during proton exchange reactions in the gas phase is about 12.0 kcal/mol, which is a considerable challenge at the atmospheric temperature. However, the possibility of hydrogen transfer to the surrounding OH can change significantly in the presence of condensed water droplets, which will be examined later.

Several hydrogen transfers were detected during our MD simulations on the OH* in condensed droplets. Therefore, it can be expected that such successive hydrogen transfers done in condensed droplets will consume less energy than those performed in the gas phase. We conducted a constrained MD calculation to calculate the energy required for these hydrogen transfers. Using the difference between the O–H and O*–H distances as the constraint variable R, slow-growth free energy calculations [45] were performed to quantify the free energy barrier of transferring hydrogen from neighboring water molecules to OH*. Henceforth, O and H will represent oxygen and hydrogen atoms obtained from water molecules, respectively, while O* and H* will indicate

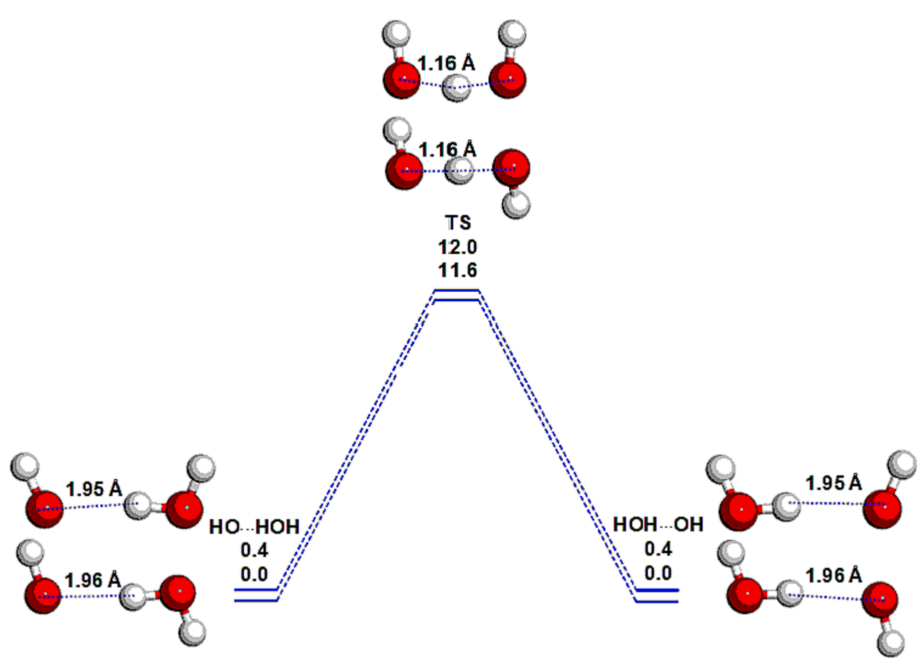

Fig. 4. Potential energy profile (kcal/mol) of the reaction between OH* and H_2O in the gas phase.

those originating from the hydroxyl radical. The average constraint force was measured over a 4.8 picosecond trajectory, employing a growth rate of 0.00025 Å/fs. The free energy profile was derived through a thermodynamic integration performed over the specified coordinate, R. As a result of the calculation, the calculated energy barrier was determined to be about 3.5 kcal/mol, Fig. 5**,** indicating that the possibility of hydrogen transfer in condensed droplets is more significant than in the gas phase (12.0 kcal/mol).

3.2. The radial distribution and coordination analysis

The probability distribution of water molecules around OH* within BOMD simulations was calculated using radial distribution functions (RDF) in light of the reactivity of the OH* during interaction with condensed droplets, Fig. 6. This can yield valuable information about the distance between neighboring water molecules and OH* based on the probability distribution of finding a particle's center at a given position, r. The pair correlation function was described as $g(r) = \frac{V}{4\pi r^2 N^2}\langle \sum_i \sum_{j\neq i} \delta(r - r_{ij})\rangle$, where δ and rij are the Dirac's delta function and separation vector between radical-neighboring molecules. Here, O* and H* denote the oxygen and hydrogen atoms from hydroxyl radical, while O and H denote those from the water molecules, respectively. The results of the RDF analysis showed that the density of water molecules around the oxygen of hydroxyl radical (O*) is significantly higher than that of hydrogen (H*). On average, it can be seen that the O* makes a bond of about 1.5 Å with the hydrogen of neighboring water molecules, while the bond between the H* and the O is weaker (1.85 Å), Fig. 6**(a)**. Investigation of RDF(O*–O) intensity peaks for both S-I and S-II systems occurred at more than 2.5 Å, indicating the absence of hemibond. Note that our previous studies on the OH* at the air–water interface in different temperatures confirmed the existence of hemibond by appearing unusual RDF(O*–O) peaks around 2.3 Å [27,28,31]. However, no unique peak around 2.3 Å is detected in RDF(O*–O) in this work, demonstrating the absence of hemibond in the condensed droplets, Fig. 6**(b)**. There is a need to emphasize that the bond created between O*–O is much stronger than the bond formed between O–O in the both S-I and S-II systems. The enhanced strength of O*–O bonds compared to O–O bonds can be attributed to two factors: the existence of unpaired electrons and electron delocalization within the bonds.

We have used the coordination analysis to obtain detailed information about the number of bonds created between OH* and neighboring water molecules, Fig. 7. Using this analysis, we determined accurately how many bonds are formed between OH* and H_2O during BOMD simulation. The coordination analysis related to S-I and S-II systems shows that the hydroxyl radical has the most significant tendency to form 4 or 3 hydrogen bonds (O*–H ~ 1.8 Å) from its oxygen side with neighboring water molecules. While the formation of a hydrogen bond from the hydrogen side of the hydroxyl radical with other water molecules is usually zero or ultimately limited to one hydrogen bond (H*–O ~ 1.8 Å). This suggests that the OH* is highly solvated in water and is capable of forming strong hydrogen bonds with neighboring water molecules (O*–H). This solvation behavior may be attributed to the high polarity of the hydroxyl group, which allows it to interact strongly with the polar water molecules. Moreover, coordination analysis showed no hemibond (O*–O ~ 2.3 Å) trace was found in both S-I and S-II systems during 80 ps of BOMD calculations.

3.3. The dominant structures

Population analysis is a powerful and versatile tool that can be used for various chemical research applications. It allows us to develop more accurate models of complex chemical reactions by providing insights into the behavior of chemical systems. Herein, population analysis was performed to investigate the dominant structures of $OH^*(H_2O)_n$ during

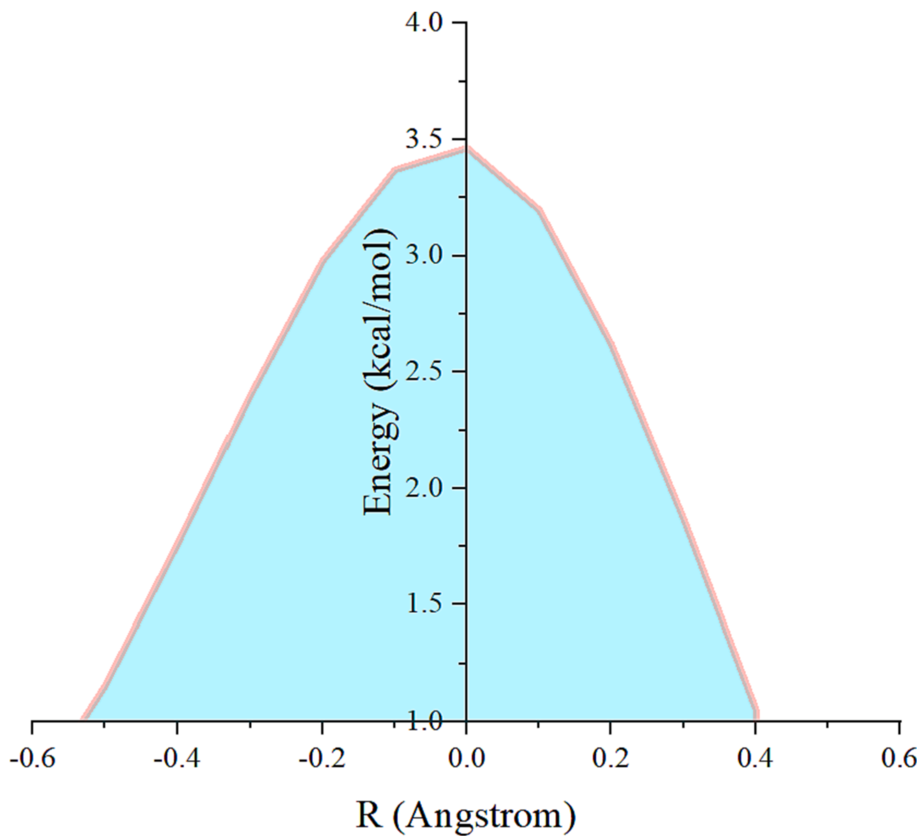

Fig. 5. The free energy barrier related to hydrogen transfer between neighboring water molecules and OH*. Where R is the difference between the O–H and O*–H bonds.

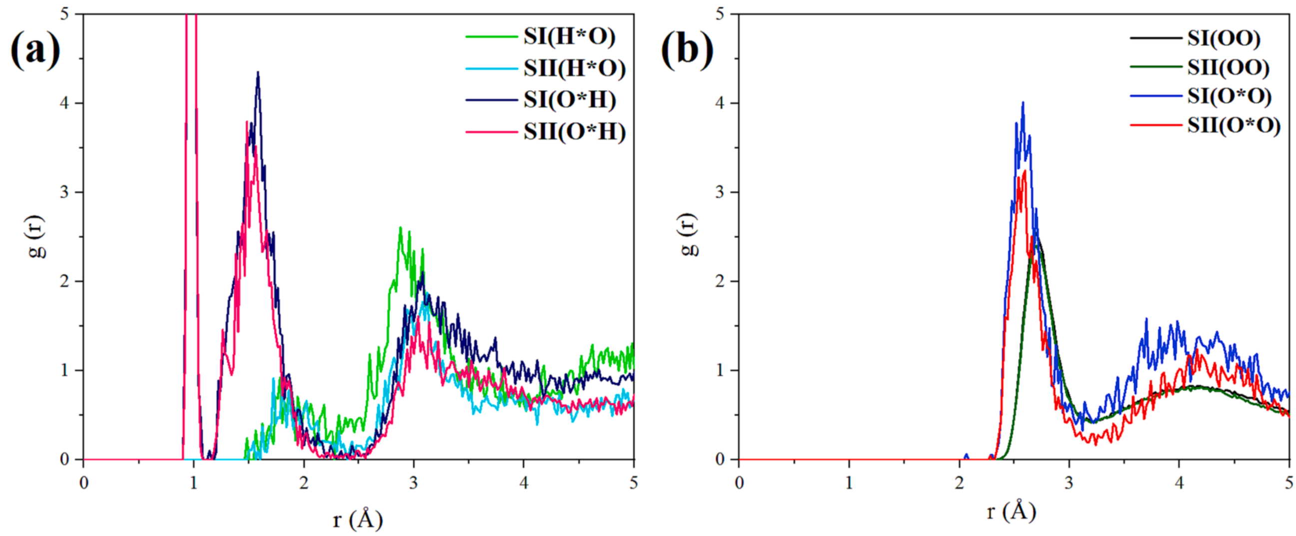

Fig. 6. Radial distribution functions for OH* and neighboring water molecules. Here, O* and H* denote the oxygen and hydrogen atoms from hydroxyl radical, while O and H denote those from the water molecules, respectively.

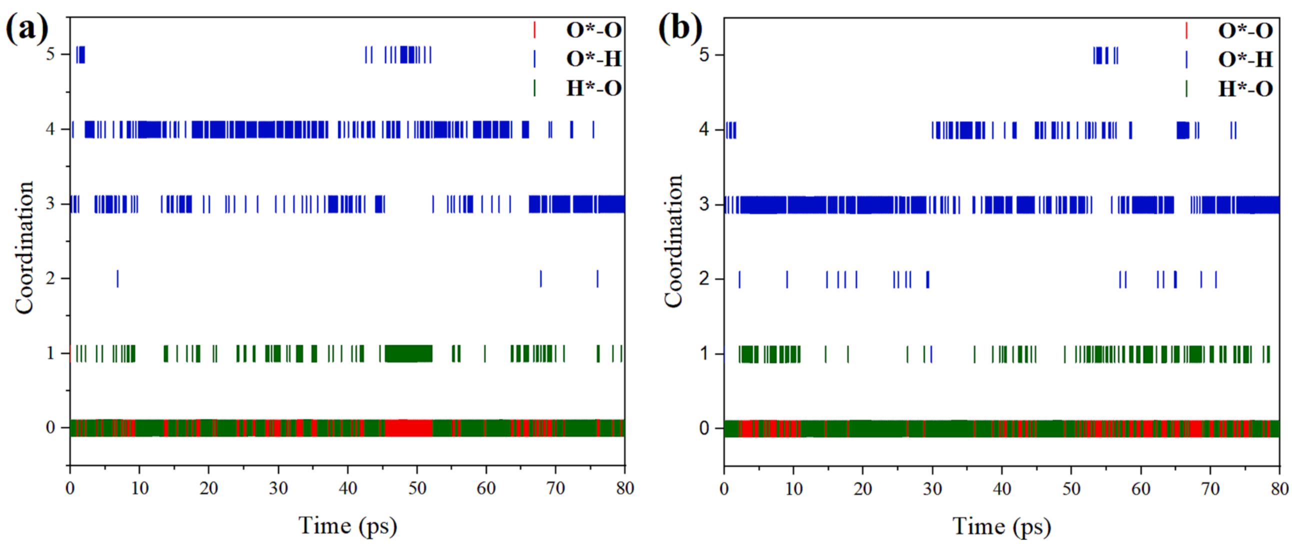

Fig. 7. The number of O*–O, O*–H and H*–O bonds created between the OH* and surrounding water molecules, related to S-I (a) and S-II (b) systems.

the interaction process between the OH* and the condensed water droplet, Fig. 8. Where the $(H_2O)_m\ldots O^*–H\ldots(H_2O)_n$ is related to configurations in which water molecules are connected to the OH* either from O* or H. As can be seen, the dominant structure in the S-I system is related to the configuration in which the OH* from its oxygen side makes four-fold coordination (40 %) with neighboring water molecules.

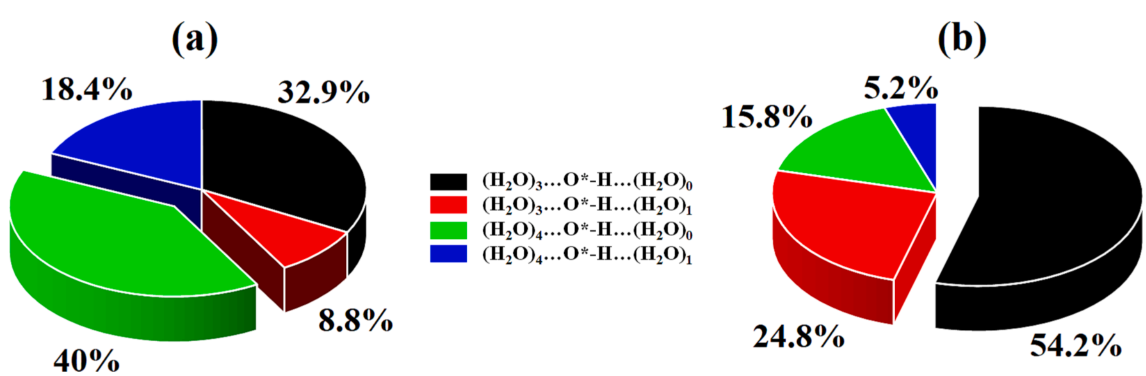

Fig. 8. The population analysis of $(H_2O)_m\ldots O^*–H\ldots(H_2O)_n$ configuration for S-I (a) and S-II(b) systems. The indices *m* and *n* represent the number of neighboring water molecules that interact with the oxygen and hydrogen components of the OH*, respectively.

In comparison, the dominant structure in the S-II system is related to the configuration in which three water molecules are attached to the oxygen side of the OH* through three-fold coordination (54.2 %). Notably, neither dominant structure in the S-I or S-II systems make hydrogen bonds between the hydrogen side of the OH* and the neighboring water molecule. The results of this study are fundamentally different from those of our previous study on the OH* in typical water droplets, where the $(H_2O)_2$…O*–H…$(H_2O)_1$ configuration along with one O*–O hemibond was detected as the most common solvation structure [27].

Angular analysis is a crucial tool in structural chemistry because it provides a way to understand the 2D arrangement of atoms within molecules. These angles can give information on the conformational stability and reactivity of OH*$(H_2O)_n$. The two-dimensional (2D) distribution functions P(r,θ) related to the first hydration shell of the OH* were computed and depicted to accurately depict the structure of hydrated OH* during its accommodation, Fig. 9. An angle H–O*-X is defined as the angle between the nearest atoms of the neighboring water molecules (X) and the hydroxyl radical, where O* is set in the center of the coordinates and H of radical is maintained along the z-axis. The results of angle analysis showed that the angle between the OH* and the nearest neighboring water molecule is between 80° and 100°, so that the distance between the OH* and neighboring water molecule is about 2.0 Å–2.5 Å in both S-I and S-II systems. Accordingly, if the hydroxy radical be aligned with the z-axis, the nearest neighboring water molecule will be perpendicular to it in the xy plane. In other words, it means that more space will be available for other water molecules to interact with the OH* simultaneously.

To fully understand the molecular stability of configurations, LUMO (lowest unoccupied molecular orbital) and spin density through the distribution of unpaired electrons are important electronic features. Upon analyzing Fig. 10, we can infer that the LUMO orbital formed around the W2 atom in the S-I system exhibits a delocalized form. This means that the electron density associated with this orbital is spread out over a larger region, rather than being localized solely around the W2 atom. This delocalization has significant consequences for the neighboring W3 atom. The delocalized LUMO orbital of W2 interacts with the orbitals of W3, causing an overlap between them. This orbital overlap between W2 and W3 leads to a phenomenon known as stabilization, wherein the overall energy of the system is lowered. In other words, the interaction between the orbitals of W2 and W3 results in a more favorable and stable molecular configuration. Similarly, a similar interpretation can be applied to the S-II system. In this case, the orbital interaction between W1 and W3, along with their overlapping with W2, also contributes to the stability of the S-II configuration. According to the spin density analysis, the orbitals are symmetrical and the electrons have a symmetric distribution around OH*, proving the absence of hemibond in the condensed system. Our previous studies showed that the spin density orbitals reveal delocalized distribution around OH* at various temperatures due to the presence of hemibonds, resulting in an asymmetric distribution of electrons [27,28,31].

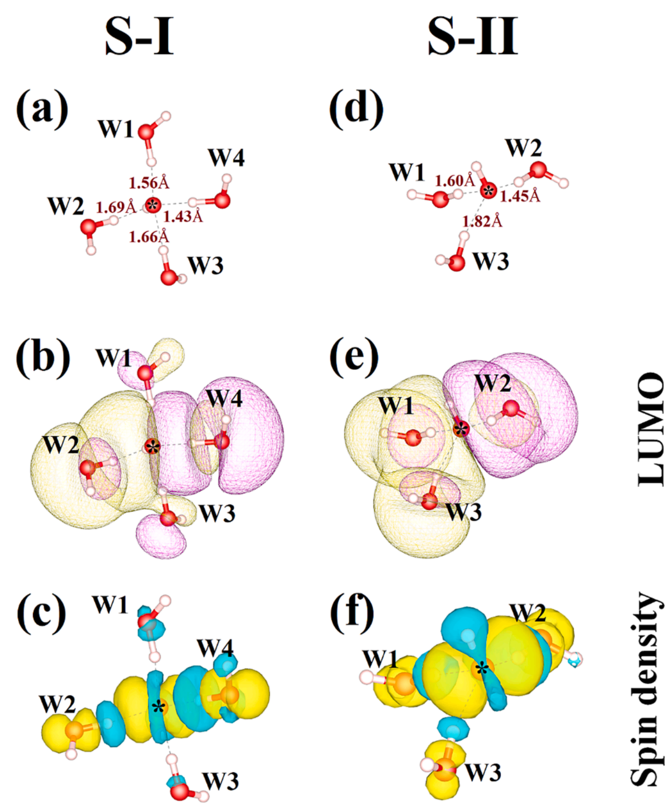

Fig. 10. A snapshot of OH* during interactions with neighboring water molecules in S-I and S-II systems: (a,d) intermolecular distance, (b,e) the lowest unoccupied molecular orbital (LUMO), (c,f) spin density distribution. Where W1, W2, W3, and W4 indicate neighboring water molecules.

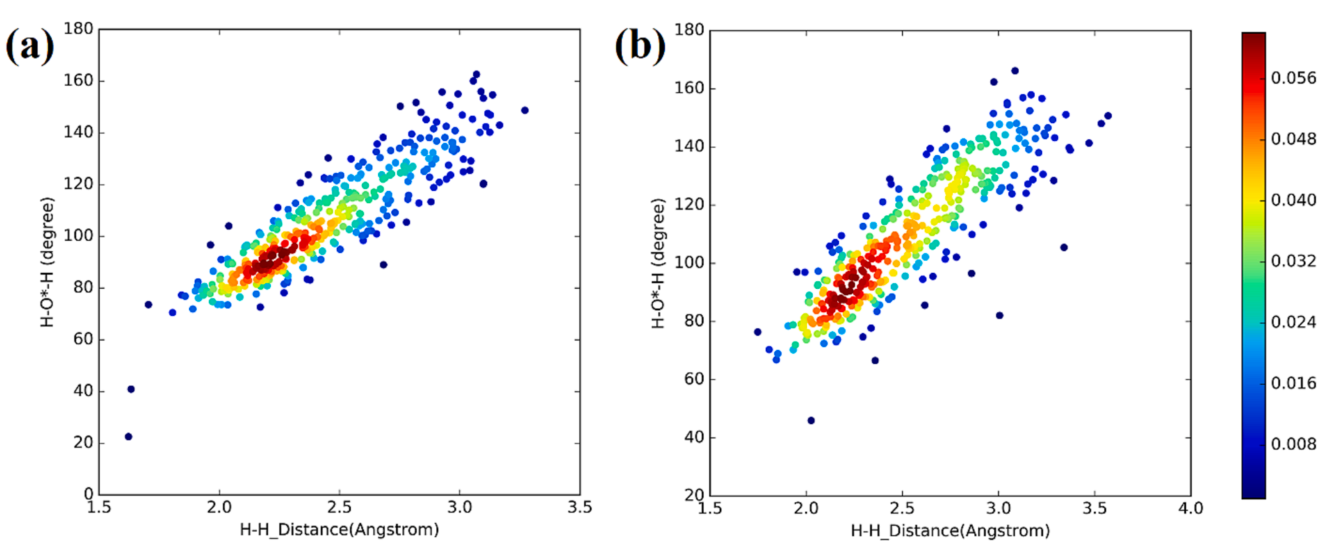

Fig. 9. Angular analysis related to dominant structures in S-I (a) and S-II(b) systems. The notation H-O*–H is used to represent the angle between the oxygen atom of OH* and its nearest neighboring water molecule. Additionally, H-H indicates the bond formed between the hydrogen atoms of OH* and the nearest neighboring water molecule. The red area corresponds to the region that indicates the most probable state.

4. Conclusion

In this study, we investigated the behavior of the OH* in interactions with condensed water droplets. The results confirmed that the OH* tends to go to the outer layers but not the droplet surface. Hydroxyl radical mobility was determined by successive hydrogen transfers with a barrier energy of about 3.5 kcal/mol, which indicates hydrogen transfer is much more likely in condensed droplets. The radial distribution function and coordination analysis results showed that the dominant structures in S-I and S-II systems are related to the configurations in which the oxygen of the OH* establishes four-fold coordination (40 %) and three-fold coordination (54.2 %) with neighboring water molecules, respectively. As a matter of fact, the OH* is not likely to form hydrogen bonds from its hydrogen side with other water molecules. In contrast to our previous investigations at various temperatures, spin density analysis did not reveal hemibonds in this study. Furthermore, our simulations showed that the presence of condensed droplets significantly affects the mobility of OH* during its interaction with water nanodroplets. In fact, the collision density of the condensed droplets may have hampered OH's mobility, resulting in an accommodation zone below the air–water interface. Despite the difficulties of measuring the OH* reactivity in condensed droplet systems and its short lifetime, experimental verification can now be carried out by utilizing the simulation results from the present study as a basis for experimental validation.

CRediT authorship contribution statement

Mohammad Hassan Hadizadeh: Formal analysis, Methodology, Investigation. **Fei Xu:** Validation, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

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